

Myriad CLOB Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

Immeas

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Myriad on Myriad CLOB. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated March 13, 2026, was delivered to Myriad, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 6 days, the Cyfrin team conducted an audit on the Myriad CLOB code provided by Myriad. The findings consist of 2 High, 1 Medium & 7 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Myriad CLOB
Repository	polkamarkets-js
Commit	b14e782840e2...
Fix Commit	566f20e22ba5...
Audit Timeline	Mar 5th - Mar 12th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	2	2	0
Medium Risk	1	1	0
Low Risk	7	7	0
Informational	14	11	3
Gas Optimizations	4	4	0
Total	28	25	3

6 Audit Scope

The audit scope was limited to:

```
contracts/AdminRegistry.sol
contracts/ConditionalTokens.sol
contracts/FeeModule.sol
contracts/MyriadCTFExchange.sol
contracts/NegRiskAdapter.sol
contracts/PredictionMarketV3ManagerCLOB.sol
contracts/WrappedCollateral.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Myriad is a binary prediction market platform built on BSC. Users trade outcome tokens representing Yes/No positions on real-world events. Markets are settled by pluggable oracle contracts. The exchange operates as a Central Limit Order Book (CLOB), matching signed off-chain orders on-chain.

8.1 Core Contracts

AdminRegistry manages the protocol's access-control roles and is built on OpenZeppelin AccessControl.

PredictionMarketV3ManagerCLOB is the market lifecycle registry. It stores market metadata, collateral token, close time, oracle, fee module and resolution outcome, and exposes the `IMyriadMarketManager` interface consumed by other contracts.

ConditionalTokens is an ERC-1155 contract where each outcome position is a token ID derived from `(marketId << 1) | outcomeIndex`. Users call `splitPosition` to deposit collateral and receive equal amounts of outcome-0 and outcome-1 tokens. After resolution, winners call `redeemPosition`; voided markets use `redeemVoided`.

MyriadCTFExchange is the on-chain order book settlement layer. It receives signed maker/taker order pairs, validates signatures, checks fill amounts, deducts fees via the market's fee module, transfers outcome tokens between counterparties, and records filled amounts in `filledAmounts[orderHash]`. Orders may be partially filled across multiple transactions.

FeeModule stores tiered fee schedules per market (up to 100 tiers, sorted ascending by price). The exchange transfers collateral to FeeModule and calls `accrueFees`; accrued fees are later withdrawn to the configured treasury address.

NegRiskAdapter enables "neg-risk" event markets. A NegRisk event groups up to 255 binary outcome markets under a single `eventId`, and each outcome market uses a shared WrappedCollateral token. Traders hold positions in individual outcome markets; since exactly one outcome resolves YES, holding a losing position in all other markets is equivalent to holding the winning position. `convertPosition` lets users exchange tokens across outcome markets within the same event.

WrappedCollateral is an ERC-20 wrapper around a base collateral token with 1:1 deposit and withdrawal. It serves as the collateral token for NegRisk event markets and is minted and burned by the NegRiskAdapter.

8.2 Market Types

Standard binary markets use any ERC-20 as collateral, resolve to outcome 0 or 1 (or -1 for voided), and are fully independent.

NegRisk event markets group multiple correlated YES/NO questions (e.g. "Which team wins the championship?"). Each question gets its own binary market. All outcome markets in an event share a WrappedCollateral token minted by the NegRiskAdapter, and exactly one question is expected to resolve YES.

8.3 Order Matching

The exchange supports two match types:

- **Standard match** (`matchOrders`): a single maker and taker trade in the same market.
- **Cross-market match** (`matchCrossMarketOrders`): two makers trade across different markets within the same NegRisk event, with no explicit taker. This allows traders to effectively swap positions across correlated markets.

All orders are signed off-chain (EIP-712). An order includes: maker address, token IDs for what is sold and bought, amounts, nonce, and an optional signature expiry.

8.4 Collateral Flow

For standard markets, collateral lives in ConditionalTokens for the duration of open positions. On split, collateral is locked 1:1 against the pair of outcome tokens minted. On merge or redeem, collateral is returned. Fees are denominated in the outcome token being sold (not the underlying collateral) and transferred to FeeModule at match time.

For NegRisk markets, collateral is wrapped into WrappedCollateral by the adapter. The adapter mints an additional "backing" position whenever a user splits, so that the last standing outcome has enough collateral to pay out all holders.

9 Attestation

Cyfrin attests that it performed a security review of the Myriad CLOB code provided by Myriad.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-03-13-cyfrin-myriad-clob-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Myriad has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=ea1838c6672b7bdef6fcfb3d7c9953c77c271520c92910f25ef4df3e7799618b
F=2026-03-13-cyfrin-myriad-clob-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.